Chapter 8

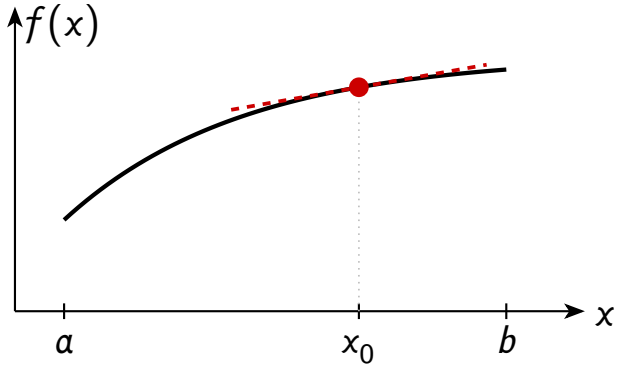
Numerical differentiation

Numerical differentiation - goal

Goal: given $f(x)$, $x \in [a, b]$, compute the derivative of f at a point x_0

Analytically: limit of the incremental ratio

$$\left.\frac{\mathrm{d}f}{\mathrm{d}x}\right|_{x_0} = \lim_{\Delta x \to 0} \frac{f(x_0 + \Delta x) - f(x_0)}{\Delta x}$$



Geometrically: slope of the tangent to f at x_0 (red, dashed)

The limit uses f at points **arbitrarily close** to x_0: f known for every x, e.g. as a formula

Numerical differentiation - nodal values

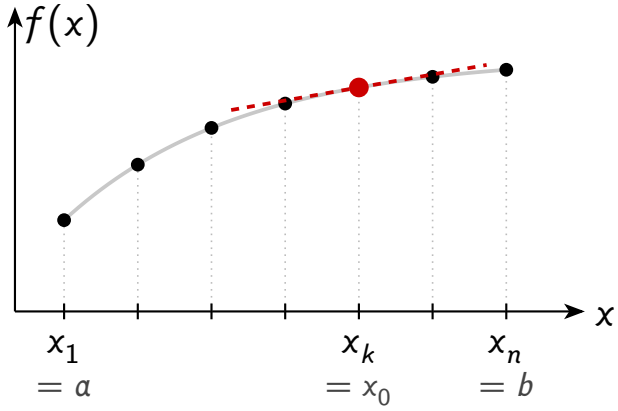
Numerically: f is known only at a **finite** number of points, e.g. the unknown of a field problem computed on a grid

Subdividing $[a, b]$ with n **nodes** ($n-1$ intervals):

$$x_1 = a, \quad x_2, \quad \ldots, \quad x_k = x_0, \quad \ldots, \quad x_n = b$$

Nodal values: the values of f at the nodes, $f_k = f(x_k)$, $k = 1, \ldots, n$

gray curve: f between the nodes is not available



Idea: estimate the derivative at x_k using only the nodal values $f_1, f_2, \ldots, f_n$

Taylor expansion

First derivative

Second derivative

Numerical differentiation - from nodal values to the derivative

Question: the nodal values $f_{k-1}, f_k, f_{k+1}, \ldots$ are just numbers. How do we get the derivative at x_k from them?

The derivative is not among the nodal values: we need a relation between

- the value of f at a **neighbouring** node, e.g. f_{k+1}
- the value **and the derivatives** of f at x_k

Such a relation exists: the **Taylor expansion** of f around x_k

Taylor expansion - series

Taylor expansion of $f(x)$ around x_0

$$f(x) = f(x_0) + \left.\frac{\mathrm{d}f}{\mathrm{d}x}\right|_{x_0} (x - x_0) + \left.\frac{1}{2}\frac{\mathrm{d}^2 f}{\mathrm{d}x^2}\right|_{x_0} (x - x_0)^2 + \left.\frac{1}{6}\frac{\mathrm{d}^3 f}{\mathrm{d}x^3}\right|_{x_0} (x - x_0)^3 + \ldots$$

Note: infinitely many terms; coefficient of the power m: $1/m!$

Truncating after the second-order term:

$$f(x) = f(x_0) + \left.\frac{\mathrm{d}f}{\mathrm{d}x}\right|_{x_0} (x - x_0) + \left.\frac{1}{2}\frac{\mathrm{d}^2 f}{\mathrm{d}x^2}\right|_{x_0} (x - x_0)^2 + O\Big((x - x_0)^3\Big)$$

$O(\ldots)$: "big-O notation"

Taylor expansion - big-O notation

$O\left((x-x_0)^3\right)$ stands for the error $E(x)$ introduced by truncating the Taylor series

$$|E(x)| = \left| f(x) - \left[f(x_0) + \left.\frac{\mathrm{d}f}{\mathrm{d}x}\right|_{x_0} (x-x_0) + \frac{1}{2}\left.\frac{\mathrm{d}^2 f}{\mathrm{d}x^2}\right|_{x_0} (x-x_0)^2 \right] \right|$$

$E(x) = O\left((x-x_0)^3\right)$ means:

$$\exists M \in \mathbb{R}^+ : \quad |E(x)| \leq M\ |x-x_0|^3 \quad \text{for } (x-x_0) \to 0$$

"there exists a positive real number M such that, for x close to x_0, $|E(x)|$ is at most $M\ |x-x_0|^3$"

Error $O\left((x-x_0)^N\right)$: goes to zero as $(x-x_0)^N$ for $x \to x_0$

Taylor expansion - halving the distance

Question: error $O\left((x-x_0)^N\right)$. What happens if the distance $(x-x_0)$ is halved?

Distance $(x-x_0)$: error $E \approx M\ |x-x_0|^N$; new point x' with $(x'-x_0) = \frac{1}{2}(x-x_0)$:

$$E' \approx M\ |\frac{1}{2}(x-x_0)|^N$$

$$\downarrow\ |1/2\ (x-x_0)|^N = (1/2)^N\ |x-x_0|^N$$

$$= \left(\frac{1}{2}\right)^N M\ |x-x_0|^N \approx \left(\frac{1}{2}\right)^N E$$

Example, $N=2$: $E' \approx (1/2)^2 E = E/4$

> Error $O\left((x-x_0)^N\right)$: halving the distance divides the error by 2^N

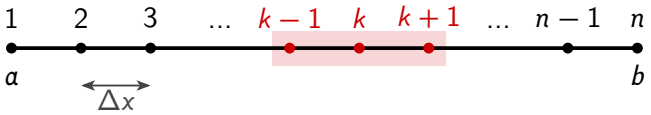
Taylor expansion

First derivative

Second derivative

Finite differences - Taylor expansion between nodes

Hp: uniform spacing, $x_{k+1} - x_k = x_k - x_{k-1} = \Delta x$



Taylor expansion around $x_0 = x_k$, evaluated at the next node $x = x_{k+1}$:

$$f(x) = f(x_0) + \left.\frac{\mathrm{d}f}{\mathrm{d}x}\right|_{x_0} (x - x_0) + \left.\frac{1}{2}\frac{\mathrm{d}^2 f}{\mathrm{d}x^2}\right|_{x_0} (x - x_0)^2 + O\left((x - x_0)^3\right)$$

$$\downarrow \; x = x_{k+1},\ x_0 = x_k \colon x - x_0 = \Delta x,\ f(x_{k+1}) = f_{k+1},\ f(x_k) = f_k$$

$$f_{k+1} = f_k + \left.\frac{\mathrm{d}f}{\mathrm{d}x}\right|_k \Delta x + \left.\frac{1}{2}\frac{\mathrm{d}^2 f}{\mathrm{d}x^2}\right|_k \Delta x^2 + O(\Delta x^3) \quad [1]$$

$\ldots|_k$: derivative at the node x_k

Finite differences - forward formula: truncation

Starting from the expansion [1] to the next node

$$f_{k+1} = f_k + \left.\frac{\mathrm{d}f}{\mathrm{d}x}\right|_k \Delta x + \left.\frac{1}{2}\frac{\mathrm{d}^2 f}{\mathrm{d}x^2}\right|_k \Delta x^2 + O(\Delta x^3)$$

$$\downarrow \text{ truncating: } 1/2\ \mathrm{d}^2 f/\mathrm{d}x^2\ \Delta x^2 + O(\Delta x^3) = O(\Delta x^2)$$

$$f_{k+1} = f_k + \left.\frac{\mathrm{d}f}{\mathrm{d}x}\right|_k \Delta x + O(\Delta x^2)$$

$$\downarrow \text{ subtracting } f_k \text{ from both members}$$

$$f_{k+1} - f_k = \left.\frac{\mathrm{d}f}{\mathrm{d}x}\right|_k \Delta x + O(\Delta x^2)$$

the second-order term is now part of the error

Finite differences - forward formula

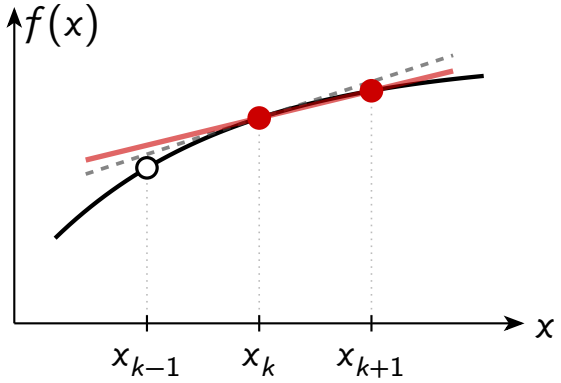
$$f_{k+1} - f_k = \left.\frac{\mathrm{d}f}{\mathrm{d}x}\right|_k \Delta x + O(\Delta x^2)$$

$$\downarrow \text{ dividing by } \Delta x$$

$$\frac{f_{k+1} - f_k}{\Delta x} = \left.\frac{\mathrm{d}f}{\mathrm{d}x}\right|_k + \frac{O(\Delta x^2)}{\Delta x}$$

$$\downarrow O(\Delta x^2)/\Delta x = O(\Delta x)$$

$$\frac{f_{k+1} - f_k}{\Delta x} = \left.\frac{\mathrm{d}f}{\mathrm{d}x}\right|_k + O(\Delta x)$$



red: secant, forward formula

dashed: tangent (exact)

Reordering:

$$\left.\frac{\mathrm{d}f}{\mathrm{d}x}\right|_k = \frac{f_{k+1} - f_k}{\Delta x} + O(\Delta x)$$

forward finite difference: **first-order** accurate

Finite differences - expansion to the previous node

Same expansion around $x_0 = x_k$, now evaluated at the **previous** node $x = x_{k-1}$

$$f(x) = f(x_0) + \left.\frac{\mathrm{d}f}{\mathrm{d}x}\right|_{x_0} (x - x_0) + \left.\frac{1}{2}\frac{\mathrm{d}^2 f}{\mathrm{d}x^2}\right|_{x_0} (x - x_0)^2 + O\Big((x - x_0)^3\Big)$$

$$\downarrow \; x = x_{k-1},\ x_0 = x_k : x - x_0 = -\Delta x$$

$$f_{k-1} = f_k + \left.\frac{\mathrm{d}f}{\mathrm{d}x}\right|_{k} (-\Delta x) + \left.\frac{1}{2}\frac{\mathrm{d}^2 f}{\mathrm{d}x^2}\right|_{k} (-\Delta x)^2 + O((-\Delta x)^3)$$

$$\downarrow \; (-\Delta x)^2 = \Delta x^2, |-\Delta x|^3 = \Delta x^3$$

$$f_{k-1} = f_k - \left.\frac{\mathrm{d}f}{\mathrm{d}x}\right|_{k} \Delta x + \left.\frac{1}{2}\frac{\mathrm{d}^2 f}{\mathrm{d}x^2}\right|_{k} \Delta x^2 + O(\Delta x^3) \quad [2]$$

odd powers of $(-\Delta x)$ change sign, even powers do not

Finite differences - backward formula

Same steps as for the forward formula, starting from [2]:

$$f_{k-1} = f_k - \left.\frac{\mathrm{d}f}{\mathrm{d}x}\right|_k \Delta x + \left.\frac{1}{2}\frac{\mathrm{d}^2 f}{\mathrm{d}x^2}\right|_k \Delta x^2 + O(\Delta x^3)$$

$$\downarrow \text{ truncating: } 1/2\ \mathrm{d}^2 f/\mathrm{d}x^2\ \Delta x^2 + O(\Delta x^3) = O(\Delta x^2)$$

$$f_{k-1} = f_k - \left.\frac{\mathrm{d}f}{\mathrm{d}x}\right|_k \Delta x + O(\Delta x^2)$$

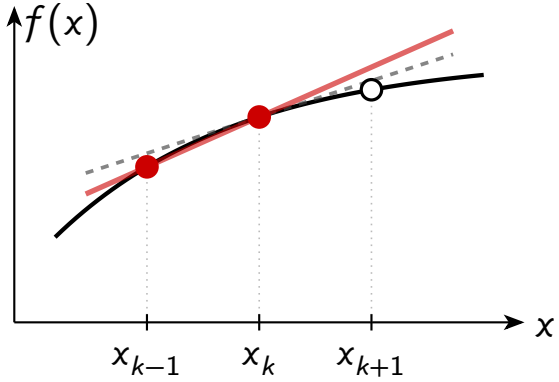
Reordering:

$$\left.\frac{\mathrm{d}f}{\mathrm{d}x}\right|_k \Delta x = f_k - f_{k-1} + O(\Delta x^2)$$

Dividing by Δx, $O(\Delta x^2)/\Delta x = O(\Delta x)$:

$$\left.\frac{\mathrm{d}f}{\mathrm{d}x}\right|_k = \frac{f_k - f_{k-1}}{\Delta x} + O(\Delta x)$$

backward finite difference: **first-order** accurate



red: secant, backward formula

dashed: tangent (exact)

Finite differences - towards second order

Question: forward and backward formulas are first order, $O(\Delta x)$. Can the nodal values give a **second-order** formula, $O(\Delta x^2)$?

Origin of the $O(\Delta x)$ error: the second-derivative term of [1] (or [2]), moved into the error and divided by Δx

Comparing [1] and [2], term by term:

$$f_{k+1} = f_k + \left.\frac{\mathrm{d}f}{\mathrm{d}x}\right|_k \Delta x + \left.\frac{1}{2}\frac{\mathrm{d}^2 f}{\mathrm{d}x^2}\right|_k \Delta x^2 + O(\Delta x^3) \quad [1]$$

$$f_{k-1} = f_k - \left.\frac{\mathrm{d}f}{\mathrm{d}x}\right|_k \Delta x + \left.\frac{1}{2}\frac{\mathrm{d}^2 f}{\mathrm{d}x^2}\right|_k \Delta x^2 + O(\Delta x^3) \quad [2]$$

first derivative: opposite signs second derivative: same sign

Idea: subtracting [2] from [1], the second-order term cancels and the first-order term doubles

Finite differences - centered formula: subtracting

$$f_{k+1} = f_k + \left.\frac{\mathrm{d}f}{\mathrm{d}x}\right|_k \Delta x + \left.\frac{1}{2}\frac{\mathrm{d}^2 f}{\mathrm{d}x^2}\right|_k \Delta x^2 + O(\Delta x^3) \quad [1]$$

$$f_{k-1} = f_k - \left.\frac{\mathrm{d}f}{\mathrm{d}x}\right|_k \Delta x + \left.\frac{1}{2}\frac{\mathrm{d}^2 f}{\mathrm{d}x^2}\right|_k \Delta x^2 + O(\Delta x^3) \quad [2]$$

Subtracting member by member, [1] − [2], term by term:

$$f_{k+1} - f_{k-1} = \cancel{(f_k - f_k)} + \left(\left.\frac{\mathrm{d}f}{\mathrm{d}x}\right|_k \Delta x + \left.\frac{\mathrm{d}f}{\mathrm{d}x}\right|_k \Delta x \right) + \cancel{\left(\left.\frac{1}{2}\frac{\mathrm{d}^2 f}{\mathrm{d}x^2}\right|_k \Delta x^2 - \left.\frac{1}{2}\frac{\mathrm{d}^2 f}{\mathrm{d}x^2}\right|_k \Delta x^2 \right)} + O(\Delta x^3) - O(\Delta x^3)$$

$O(\Delta x^3) - O(\Delta x^3) = O(\Delta x^3)$: the error terms do **not** cancel

$$f_{k+1} - f_{k-1} = 2\left.\frac{\mathrm{d}f}{\mathrm{d}x}\right|_k \Delta x + O(\Delta x^3)$$

Finite differences - centered formula

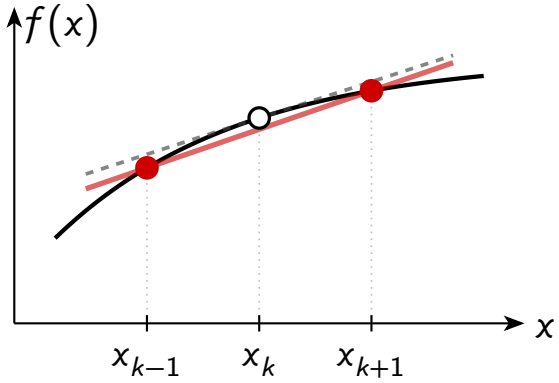
$$f_{k+1} - f_{k-1} = 2\left.\frac{\mathrm{d}f}{\mathrm{d}x}\right|_k \Delta x + O(\Delta x^3)$$

$$\downarrow \text{ dividing by } 2\Delta x$$

$$\frac{f_{k+1} - f_{k-1}}{2\Delta x} = \left.\frac{\mathrm{d}f}{\mathrm{d}x}\right|_k + \frac{O(\Delta x^3)}{2\Delta x}$$

$$\downarrow O(\Delta x^3)/(2\Delta x) = O(\Delta x^2)$$

$$\frac{f_{k+1} - f_{k-1}}{2\Delta x} = \left.\frac{\mathrm{d}f}{\mathrm{d}x}\right|_k + O(\Delta x^2)$$



red: secant, centered formula

dashed: tangent (exact)

Reordering:

$$\left.\frac{\mathrm{d}f}{\mathrm{d}x}\right|_k = \frac{f_{k+1} - f_{k-1}}{2\Delta x} + O(\Delta x^2)$$

centered finite difference: **second-order** accurate

Finite differences - one-sided second-order formula

Question: a **second-order** formula using only nodes on **one side** of x_k? (e.g. at the boundary node $x_1 = a$: no node on its left)

Idea: one more node, x_{k+2}, and a combination of two expansions that cancels the second-order term

Expansion around x_k, evaluated at $x = x_{k+2}$:

$$f_{k+2} = f_k + \left.\frac{\mathrm{d}f}{\mathrm{d}x}\right|_k (2\Delta x) + \left.\frac{1}{2}\frac{\mathrm{d}^2 f}{\mathrm{d}x^2}\right|_k (2\Delta x)^2 + O((2\Delta x)^3)$$

$$\downarrow \quad x - x_0 = 2\Delta x\text{: } (2\Delta x)^2 = 4\Delta x^2,\ O((2\Delta x)^3) = O(\Delta x^3)$$

$$f_{k+2} = f_k + 2\left.\frac{\mathrm{d}f}{\mathrm{d}x}\right|_k \Delta x + 2\left.\frac{\mathrm{d}^2 f}{\mathrm{d}x^2}\right|_k \Delta x^2 + O(\Delta x^3) \quad [3]$$

Finite differences - one-sided formula: combining

$$f_{k+1} = f_k + \left.\frac{\mathrm{d}f}{\mathrm{d}x}\right|_k \Delta x + \frac{1}{2}\left.\frac{\mathrm{d}^2 f}{\mathrm{d}x^2}\right|_k \Delta x^2 + O(\Delta x^3) \quad [1]$$

$$f_{k+2} = f_k + 2\left.\frac{\mathrm{d}f}{\mathrm{d}x}\right|_k \Delta x + 2\left.\frac{\mathrm{d}^2 f}{\mathrm{d}x^2}\right|_k \Delta x^2 + O(\Delta x^3) \quad [3]$$

second-order coefficients: $1/2$ and $2 \Rightarrow$ multiplying [1] by 4 makes them equal

Multiplying [1] by 4 ($4 \cdot O(\Delta x^3) = O(\Delta x^3)$):

$$4f_{k+1} = 4f_k + 4\left.\frac{\mathrm{d}f}{\mathrm{d}x}\right|_k \Delta x + 2\left.\frac{\mathrm{d}^2 f}{\mathrm{d}x^2}\right|_k \Delta x^2 + O(\Delta x^3)$$

Subtracting [3] member by member, term by term:

$$4f_{k+1} - f_{k+2} = (4f_k - f_k) + \left(4\left.\frac{\mathrm{d}f}{\mathrm{d}x}\right|_k \Delta x - 2\left.\frac{\mathrm{d}f}{\mathrm{d}x}\right|_k \Delta x\right) + \cancel{\left(2\left.\frac{\mathrm{d}^2 f}{\mathrm{d}x^2}\right|_k \Delta x^2 - 2\left.\frac{\mathrm{d}^2 f}{\mathrm{d}x^2}\right|_k \Delta x^2\right)} + O(\Delta x^3) - O(\Delta x^3)$$

Finite differences - one-sided forward formula

$$4f_{k+1} - f_{k+2} = 3f_k + 2\left.\frac{\mathrm{d}f}{\mathrm{d}x}\right|_k \Delta x + O(\Delta x^3)$$

$$\downarrow \text{ subtracting } 3f_k$$

$$-3f_k + 4f_{k+1} - f_{k+2} = 2\left.\frac{\mathrm{d}f}{\mathrm{d}x}\right|_k \Delta x + O(\Delta x^3)$$

$$\downarrow \text{ dividing by } 2\Delta x,\ O(\Delta x^3)/(2\Delta x) = O(\Delta x^2)$$

$$\frac{-3f_k + 4f_{k+1} - f_{k+2}}{2\Delta x} = \left.\frac{\mathrm{d}f}{\mathrm{d}x}\right|_k + O(\Delta x^2)$$

Reordering:

$$\left.\frac{\mathrm{d}f}{\mathrm{d}x}\right|_k = \frac{-3f_k + 4f_{k+1} - f_{k+2}}{2\Delta x} + O(\Delta x^2)$$

one-sided forward formula: three nodes x_k, x_{k+1}, x_{k+2}, **second-order** accurate

Finite differences - one-sided backward formula

Same steps with the nodes x_{k-1}, x_{k-2}: $x - x_0 = -\Delta x$ and $-2\Delta x$

$$f_{k-1} = f_k - \left.\frac{\mathrm{d}f}{\mathrm{d}x}\right|_k \Delta x + \frac{1}{2}\left.\frac{\mathrm{d}^2 f}{\mathrm{d}x^2}\right|_k \Delta x^2 + O(\Delta x^3) \quad [2]$$

$$f_{k-2} = f_k - 2\left.\frac{\mathrm{d}f}{\mathrm{d}x}\right|_k \Delta x + 2\left.\frac{\mathrm{d}^2 f}{\mathrm{d}x^2}\right|_k \Delta x^2 + O(\Delta x^3) \quad [4]$$

$4 \cdot [2] - [4]$, term by term:

$$4f_{k-1} - f_{k-2} = (4f_k - f_k) + \left(-4\left.\frac{\mathrm{d}f}{\mathrm{d}x}\right|_k \Delta x + 2\left.\frac{\mathrm{d}f}{\mathrm{d}x}\right|_k \Delta x \right) + \cancel{\left(2\left.\frac{\mathrm{d}^2 f}{\mathrm{d}x^2}\right|_k \Delta x^2 - 2\left.\frac{\mathrm{d}^2 f}{\mathrm{d}x^2}\right|_k \Delta x^2 \right)} + O(\Delta x^3) - O(\Delta x^3)$$

Subtracting $3f_k$, dividing by $-2\Delta x$, reordering:

$$\left.\frac{\mathrm{d}f}{\mathrm{d}x}\right|_k = \frac{3f_k - 4f_{k-1} + f_{k-2}}{2\Delta x} + O(\Delta x^2)$$

one-sided backward formula: three nodes x_k, x_{k-1}, x_{k-2}, **second-order** accurate

Numerical differentiation

Taylor expansion

First derivative

Second derivative

Finite differences - second derivative: expansions

Question: a formula for $\mathrm{d}^2 f / \mathrm{d}x^2$ at x_k from f_{k-1}, f_k, f_{k+1}?

Idea: **adding** [1] and [2], the first-derivative terms (opposite signs) cancel, the second-derivative terms add up

At the end we divide by Δx^2 (two powers lost): one more term in the expansions, error $O(\Delta x^4)$

$$f_{k+1} = f_k + \left.\frac{\mathrm{d}f}{\mathrm{d}x}\right|_k \Delta x + \frac{1}{2}\left.\frac{\mathrm{d}^2 f}{\mathrm{d}x^2}\right|_k \Delta x^2 + \frac{1}{6}\left.\frac{\mathrm{d}^3 f}{\mathrm{d}x^3}\right|_k \Delta x^3 + O(\Delta x^4) \quad [5]$$

$$f_{k-1} = f_k - \left.\frac{\mathrm{d}f}{\mathrm{d}x}\right|_k \Delta x + \frac{1}{2}\left.\frac{\mathrm{d}^2 f}{\mathrm{d}x^2}\right|_k \Delta x^2 - \frac{1}{6}\left.\frac{\mathrm{d}^3 f}{\mathrm{d}x^3}\right|_k \Delta x^3 + O(\Delta x^4) \quad [6]$$

coefficient $1/3! = 1/6$; the third power changes sign, as $(-\Delta x)^3 = -\Delta x^3$

Finite differences - second derivative: adding

Adding member by member, [5] + [6], term by term:

$$f_{k+1} + f_{k-1} = 2f_k + \cancel{\left(\left.\frac{\mathrm{d}f}{\mathrm{d}x}\right|_k \Delta x - \left.\frac{\mathrm{d}f}{\mathrm{d}x}\right|_k \Delta x \right)} + 2 \cdot \frac{1}{2} \left.\frac{\mathrm{d}^2 f}{\mathrm{d}x^2}\right|_k \Delta x^2 + \cancel{\left(\frac{1}{6} \left.\frac{\mathrm{d}^3 f}{\mathrm{d}x^3}\right|_k \Delta x^3 - \frac{1}{6} \left.\frac{\mathrm{d}^3 f}{\mathrm{d}x^3}\right|_k \Delta x^3 \right)} + O(\Delta x^4) + O(\Delta x^4)$$

$O(\Delta x^4) + O(\Delta x^4) = O(\Delta x^4)$, $2 \cdot 1/2 = 1$:

$$f_{k+1} + f_{k-1} = 2f_k + \left.\frac{\mathrm{d}^2 f}{\mathrm{d}x^2}\right|_k \Delta x^2 + O(\Delta x^4)$$

Finite differences - second derivative

$$f_{k+1} + f_{k-1} = 2f_k + \left.\frac{\mathrm{d}^2 f}{\mathrm{d}x^2}\right|_k \Delta x^2 + O(\Delta x^4)$$

$$\downarrow \text{ subtracting } 2f_k, \text{ dividing by } \Delta x^2$$

$$\frac{f_{k+1} - 2f_k + f_{k-1}}{\Delta x^2} = \left.\frac{\mathrm{d}^2 f}{\mathrm{d}x^2}\right|_k + \frac{O(\Delta x^4)}{\Delta x^2}$$

$$\downarrow O(\Delta x^4)/\Delta x^2 = O(\Delta x^2)$$

$$\left.\frac{\mathrm{d}^2 f}{\mathrm{d}x^2}\right|_k = \frac{f_{k+1} - 2f_k + f_{k-1}}{\Delta x^2} + O(\Delta x^2)$$

centered finite difference for the second derivative: **second-order** accurate

Finite differences - truncation error

Question: how does the error change when the spacing is halved?

formula	error	Δx	$\Delta x/2$
forward / backward	$O(\Delta x)$	E	$E/2$
centered (first and second derivative)	$O(\Delta x^2)$	E	$E/4$
one-sided, three nodes	$O(\Delta x^2)$	E	$E/4$

halving Δx divides an $O(\Delta x^N)$ error by 2^N

Check in the lab: errors for Δx and $\Delta x/2$ and their ratio (step 3)

Lab - Numerical derivatives (in class)

Done in class, in MATLAB

Folder: `02_numerical_derivatives`

Goal: finite difference formulas vs the exact derivative; order of accuracy, truncation and round-off; plain script `numerical_derivatives.m` (sections `%%`)

Finite differences - truncation and round-off

Question: why can't the error be made arbitrarily small by reducing $\Delta x = h$?

- **Truncation** error: finite number of Taylor terms (derivative $\rightarrow$ algebraic expression)
- **Round-off** error: finite number of digits

Forward formula in floating point: f_{k+1} and f_k carry relative errors up to ε

$$\frac{\mathrm{fl}(f_{k+1}) - \mathrm{fl}(f_k)}{h} = \frac{f_{k+1} - f_k}{h} + O(\varepsilon \, |f_k| \, / h)$$

$f_{k+1} \approx f_k$ for small h: subtraction of close numbers (cancellation)

Total error: truncation decreases with h, round-off increases as $h \rightarrow 0$

$$|E(h)| \approx \underbrace{Mh}_{\text{truncation}} + \underbrace{\varepsilon \, |f| \, / h}_{\text{round-off}}$$

Minimum where the two terms are comparable: $h \approx \sqrt{\varepsilon} \approx 10^{-8}$ for the forward formula; $h \approx \varepsilon^{1/3} \approx 10^{-5}$ for the centered one (truncation Mh^2)

higher order: smaller minimum error, reached with a larger h

Numerical differentiation - summary

- Derivatives are approximated by combinations of **nodal values**, obtained from Taylor expansions
- Forward and backward: $O(\Delta x)$; centered (first and second derivative): $O(\Delta x^2)$
- Truncation error decreases with Δx; round-off error grows as $\Delta x \to 0$
- Next step: replace the derivatives of a **differential equation** with these formulas $\to$ finite difference method

Advanced - from derivative to Jacobian

ADVANCED



Consider $\boldsymbol{f} : \mathbb{R}^n \to \mathbb{R}^m$, e.g. the residual $\boldsymbol{F}(\boldsymbol{u})$ of a discretised **nonlinear** problem, $\boldsymbol{u}$: nodal values

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Jacobian matrix $(m \times n)$: derivative of each component with respect to each variable

$$J_{ij} = \frac{\partial f_i}{\partial x_j}, \quad i = 1, \dots, m, \quad j = 1, \dots, n$$

$n = m = 1$: $J = \mathrm{d}f / \mathrm{d}x$

Taylor expansion to first order: $\boldsymbol{f}(\boldsymbol{x} + \Delta\boldsymbol{x}) = \boldsymbol{f}(\boldsymbol{x}) + J\Delta\boldsymbol{x} + O(\|\Delta\boldsymbol{x}\|^2)$

Newton method for $\boldsymbol{F}(\boldsymbol{u}) = \boldsymbol{0}$: at each iteration

$$J\left(\boldsymbol{u}^{(k)}\right) \delta\boldsymbol{u} = -\boldsymbol{F}\left(\boldsymbol{u}^{(k)}\right), \quad \boldsymbol{u}^{(k+1)} = \boldsymbol{u}^{(k)} + \delta\boldsymbol{u}$$

e.g. saturable ferromagnetic materials (μ depends on $|\boldsymbol{H}|$), temperature-dependent σ; quadratic convergence requires an accurate J

Also: **sensitivities** of an output (force, inductance) with respect to design parameters

Advanced - Jacobian by finite differences

ADVANCED



Column j of J: forward formula along the direction $\boldsymbol{e}_j$ (unit vector of variable x_j)

$$J\boldsymbol{e}_j \approx \frac{\boldsymbol{f}\left(\boldsymbol{x} + h\boldsymbol{e}_j\right) - \boldsymbol{f}\left(\boldsymbol{x}\right)}{h}, \quad j = 1, \ldots, n$$

Cost: $n + 1$ evaluations of $\boldsymbol{f}$ ($2n$ with the centered formula)

n: number of unknowns, 10^4–10^6 in a field model

Accuracy: same trade-off as in the scalar case; best relative error $\approx \sqrt{\varepsilon}$ (forward), $\approx \varepsilon^{2/3}$ (centered)

format	ε	forward, $\sqrt{\varepsilon}$	centered, $\varepsilon^{2/3}$
double	$1.1 \cdot 10^{-16}$	$1.1 \cdot 10^{-8}$	$2.3 \cdot 10^{-11}$
single	$6 \cdot 10^{-8}$	$2.4 \cdot 10^{-4}$	$1.5 \cdot 10^{-5}$
half	$4.9 \cdot 10^{-4}$	$2.2 \cdot 10^{-2}$	$6.2 \cdot 10^{-3}$

Finite difference Jacobian: $n + 1$ evaluations, about **half** the digits, h to be tuned

Advanced - automatic differentiation

ADVANCED



A program computing f is a sequence of elementary operations ($+$, $\times$, sin, exp, ...) with known derivatives

Automatic differentiation (AD): chain rule applied to each operation, on numbers

Forward mode: each variable carries the pair (value, derivative) (a, a')

$$(a, a') + (b, b') = (a + b, a' + b'), \quad (a, a') \cdot (b, b') = (ab, a'b + ab'), \quad \sin(a, a') = (\sin a, a' \cos a)$$

Example, $f(x) = x \sin x$: starting from $(x_0, 1)$

$$(x_0, 1) \cdot (\sin x_0, \cos x_0) = (x_0 \sin x_0, \quad \sin x_0 + x_0 \cos x_0)$$

Lab function at $x = 1$, with pairs: $f = 9.5$, $\mathrm{d}f/\mathrm{d}x = 19.5$ (exact: 19.5)

computed live with (value, derivative) pairs in double precision

> No h, no truncation error: derivative exact up to the round-off of the elementary operations

Advanced - automatic differentiation: cost

ADVANCED



Jacobian of $\boldsymbol{f} : \mathbb{R}^n \to \mathbb{R}^m$, one pass: a small multiple of the cost of $\boldsymbol{f}$

Forward mode: one pass per **input** direction $\boldsymbol{e}_j \to J$ column by column, n passes

Reverse mode (adjoint): one pass per **output** $\to J$ row by row, m passes

scalar output (energy, force, cost function): gradient with respect to all parameters in one pass, independent of n; intermediate values stored in memory

Sparse J (FDM, FEM: each equation involves few nodes): columns without common nonzero rows computed together (colouring)

number of passes $\approx$ nonzeros per row, instead of n; also valid for finite differences

	accuracy	passes for J	parameter h
finite differences	$\approx \sqrt{\varepsilon}$	$n+1$	yes
AD, forward	$\approx \varepsilon$	n	no
AD, reverse	$\approx \varepsilon$	m	no

Advanced - AD tools and thesis activity

ADVANCED



Implementations:

- operator overloading (C++): autodiff, CoDiPack, CppAD, Adept, Sacado (Trilinos)
- source or compiler transformation: Tapenade, Enzyme (LLVM)
- MATLAB: `dlgradient` (Deep Learning Toolbox), automatic derivatives in the Optimization Toolbox

Thesis activity Jacobians by AD in C++ for differential-algebraic problems of non-thermal plasmas (code Merlino2d):

- transport of the plasma species coupled with Poisson's equation: differential-algebraic system
- implicit time integration: Newton method, Jacobian at each iteration
- Jacobian from AD versus finite differences: accuracy, Newton convergence, computational cost